

GENERAL NOTES & LEGENDS

A LEGEND.

- 1 REFER TO LEGEND FOR THOSE SYMBOLS ONLY MENTIONED ON ALL LAYOUT PLANS. OTHER SYMBOLS SHOWN IN LEGEND BUT NOT ACTUALLY SHOWN ON LAYOUT DRAWINGS MAY PLEASE BE IGNORED.

B ELV WORKS.

- 1 THE FOLLOWING NOTES SHALL IN GENERAL APPLY TO ALL DRAWINGS. THE INSTRUCTIONS IN THESE NOTES SHALL BE FOLLOWED UNLESS ALL ELV WORKS PERTAINING TO WIRING OF ELECTRICAL SYSTEMS SHALL STRICTLY CONFORM TO I.E.E/ NFPA LATEST EDITION OF RULES & REGULATIONS.

2 FIRE ALARM DRAWINGS

ALL ELECTRICAL DRAWINGS SHALL BE READ IN CONJUNCTION WITH BOQ, TECHNICAL SPECIFICATIONS AND IN CO-ORDINATION WITH OTHER SERVICES & DRAWINGS.

3 DIMENSIONS ON ELV LAYOUT PLANS

ALL DIMENSIONS IF SHOWN ON THE DRWAING LAYOUT PLANS ARE APPROXIMATE. THE ELECTRICAL CONTRACTOR SHALL BE RESPONSIBLE TO DO ALL NECESSARY CALCULATIONS TO ARRIVE AT THE ACTUAL DIMENSIONS/MEASUREMENTS IN CO-ORDINATION WITH ALL OTHER RELEVANT DRAWINGS OF OTHER SERVICES.

4 SHOP DRAWINGS

THE ELECTRICAL CONTRACTOR SHALL PREPARE ELECTRICAL SHOP DRAWINGS WITH ALL NECESSARY RELEVANT DETAILS. THE SHOP DRAWINGS SHALL BE SUBMITTED TO ENGINEER/CONSULTANT TENDER DRAWING BEFORE THE COMMENCEMENT OF ANY WORKS AT SITE.

5 CO-ORDINATION

THE ELECTRICAL CONTRACTOR SHALL DO ALL NECESSARY CO-ORDINATION OF ELECTRICAL WORKS AND ALLIED SYSTEMS WITH ALL OTHER SERVICES AT SITE.

6 ELECTRICAL WIRING.

ALL WIRING SHALL BE DONE IN G.I CONDUIT TO BE CONCEALED/SURFACE IN WALLS,SLABS,COLUMNS AND FLOORS OR AS SHOWN ON DRAWINGS.

- 7 THE WIRING SHOULD BE STARTED ONLY AFTER THE CONDUIT SYSTEM HAS BEEN COMPLETED AND ALL OUTLET BOXES ARE FIXED AT THEIR RESPECTIVE POSITIONS.

8 LUBRICATION.

NO LUBRICATION EXCEPT AS RECOMMENDED BY MANUFACTURER SHALL BE USED FOR PULLING OF WIRES IN CONDUITS. NO OIL OF ANY KIND SHALL BE USED.

9 EARTHING OF FACP

FACP PANEL SHALL BE PROPERLY EARTHED WITH 4mm sq PVC INSULATED WIRES OF COLOR GREEN-YELLOW.

C CONSULATANT.

- 1 CONTRACTOR TO CONSULT CONSULTANT/ENGINEER FOR ANY FURTHER CLARIFICATIONS. NO ASSUMPTIONS SHOULD BE MADE.

D TESTS.

CONTRACTOR SHALL CARRY OUT ALL ELECTRICAL TESTS AT HIS EXPENSE THROUGH LICENSED ELECTRICAL SUPERVISOR IN THE PRESENCE OF ENGINEER INCHARGE OR HIS REPRESENTATIVE.



CAPACITY OF CONDUITS

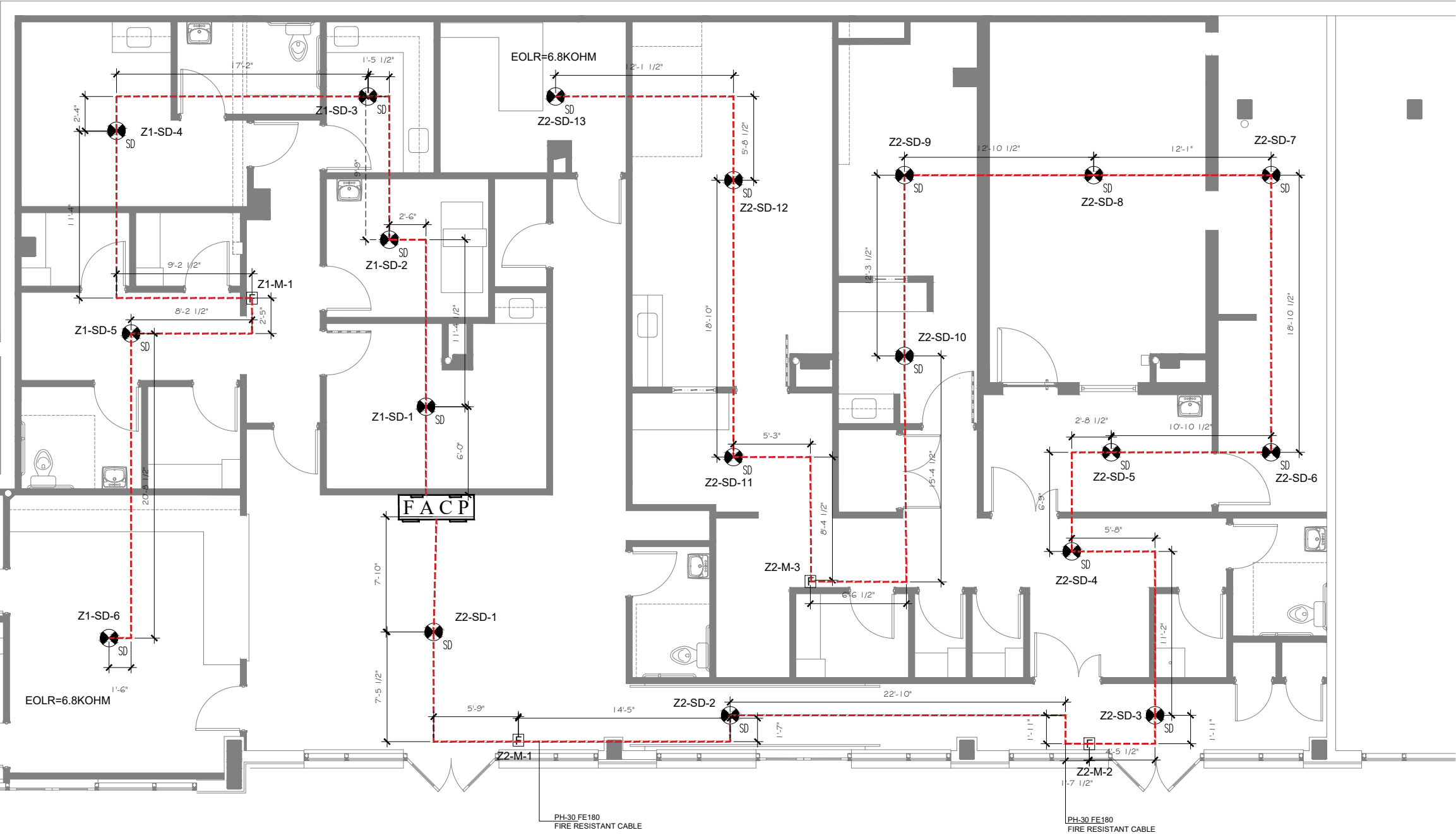
S.NO	CAPACITY OF CONDUITS			MAXIMUM NUMBER OF CABLES IN CONDUITS SIZES		
1	NOMINAL CONDUCTOR SIZE—mmSq	NO & DIA OF WIRES	OVER ALL DIAMETER	3/4” DIA (20mm DIA)	1” DIA (25mm DIA)	1—1/4” DIA (32mm DIA)
2	1.5	1/1.38	3.1	10	18	30
3	2.5	1/1.78	3.5	8	14	23
4	2.5	7/0.67	3.8	7	12	20
5	4	7/0.85	4.3	5	9	15
6	10	7/1.04	4.9	4	7	12
7	16	7/1.35	6.2	2	4	7
8	16	7/1.78	7.3	—	3	5
9	25	7/2.14	9.0	—	2	3
10	35	19/1.53	10.3	—	—	2



Zone 1 voltage drops							
2.5MMSQ fire cable per feet resistance 0.002ohm							
Sr/No	Item	Length of wire(ft) from last point	Length of wire(ft) from FACP	Resistance (ohm)	Current (340uA) per device	Voltage drop (mV)	Voltage at sensor (24V)
1	Z1-SD-1	12	12	0.024	0.00204	0.04896	23.99995104
2	Z1-SD-2	14	26	0.052	0.0017	0.0884	23.99990208
3	Z1-SD-3	11	37	0.074	0.00136	0.10064	23.99981368
4	Z1-M-1	20	57	0.114	0.00102	0.11628	23.99971304
5	Z1-SD-4	27	84	0.168	0.00102	0.17136	23.99959676
6	Z1-SD-5	16	100	0.2	0.00068	0.136	23.9994254
7	Z1-SD-6	22	122	0.244	0.00034	0.08296	23.9992894
Zone 2 voltage drops							
2.5MMSQ fire cable per feet resistance 0.002ohm							
Sr/No	Item	Length of wire(ft) from last point	Length of wire(ft) from FACP	Resistance (ohm)	Current (340uA) per device	Voltage drop (mV)	Voltage at sensor (24V)
1	Z2-SD-1	13	13	0.026	0.00442	0.11492	23.99988508
2	Z2-M-1	16	29	0.058	0.00408	0.23664	23.99964844
3	Z2-SD-2	21	50	0.1	0.00408	0.408	23.99924044
4	Z2-M-2	31	81	0.162	0.00374	0.60588	23.99863456
5	Z2-SD-3	13	94	0.188	0.00374	0.70312	23.99793144
6	Z2-SD-4	17	111	0.222	0.0034	0.7548	23.99717664
7	Z2-SD-5	10	121	0.242	0.00306	0.74052	23.99643612
8	Z2-SD-6	10	131	0.262	0.00272	0.71264	23.99572348
9	Z2-SD-7	18	149	0.298	0.00238	0.70924	23.99501424
10	Z2-SD-8	12	161	0.322	0.00204	0.65688	23.99435736
11	Z2-SD-9	13	174	0.348	0.0017	0.5916	23.99376576
12	Z2-SD-10	13	187	0.374	0.00136	0.50864	23.99325712
13	Z2-M-3	28	215	0.43	0.00102	0.4386	23.99281852
14	Z2-SD-11	14	229	0.458	0.00102	0.46716	23.99235136
15	Z2-SD-12	18	247	0.494	0.00068	0.33592	23.99201544
16	Z2-SD-13	19	266	0.532	0.00034	0.18088	23.99183456

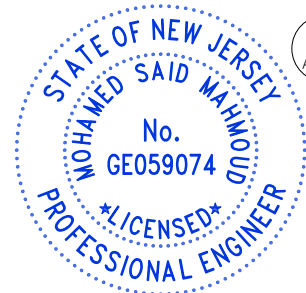
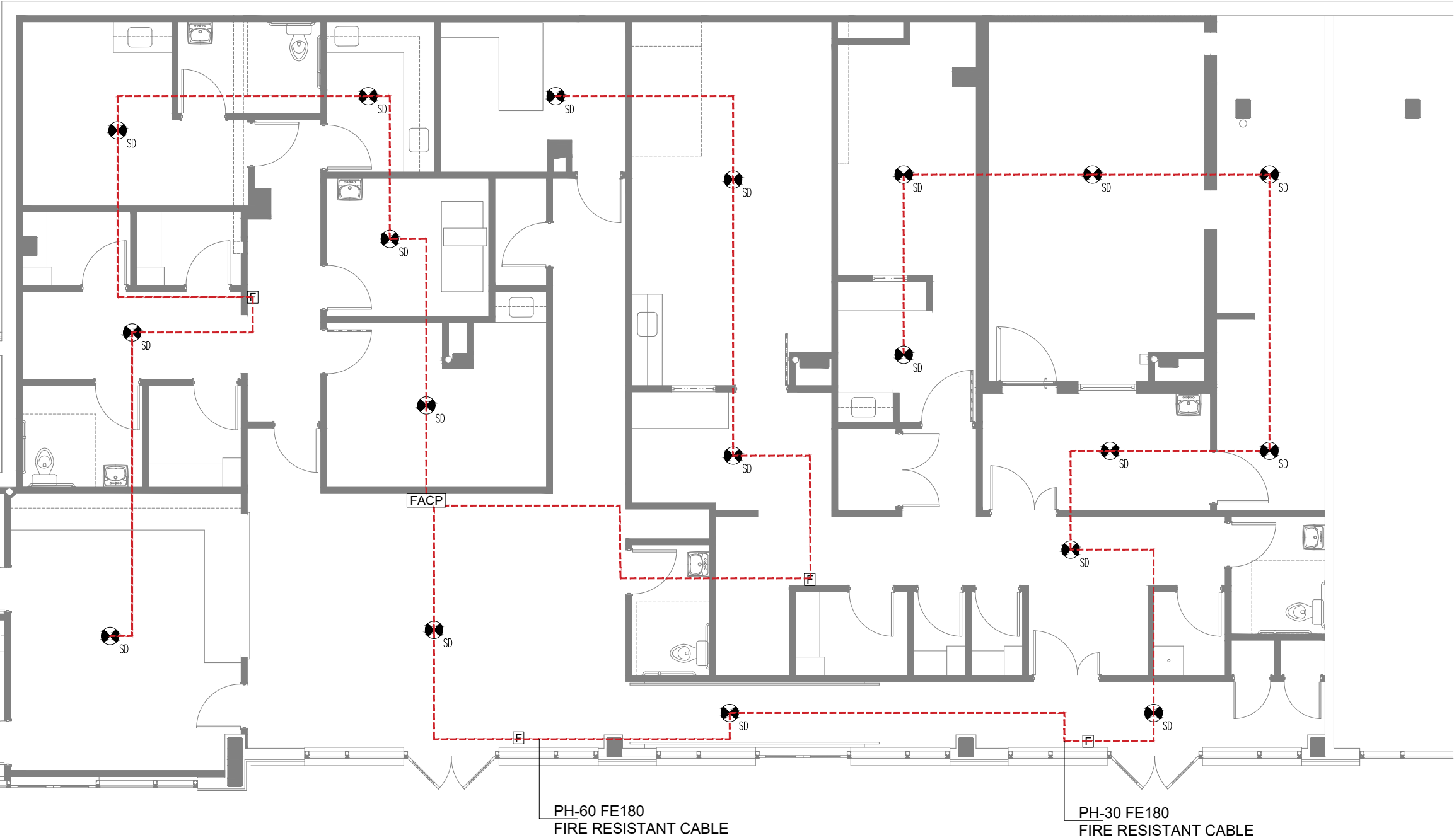


EMERGENCY SYMBOL SCHEDULE	
SYMBOL	DESCRIPTION
	EMERGENCY EXIT SIGN c/w BATTERY PACK
	EXIT SIGN WITH EMERGENCY HEADS c/w BATTERY PACK
	CHEVRON ARROW
	EMERGENCY HEADS c/w BATTERY PACK
	REMOTE EMERGENCY DOUBLE LED HEADS
	REMOTE EMERGENCY SINGLE LED HEADS
	FIRE ALARM HORN AND STROBE LIGHT
	FIRE STROBE LIGHT
	FIRE ALARM PULL STATION
	SMOKE DETECTOR
	CARBON MONOXIDE
	HEAT DETECTOR
	FIRE EXTINGUISHER
	FIRE ALARM CONTROL PANEL
	FIRE ALARM ANNUNCIATION PANEL
	REMOTE EMERGENCY HEAD
	BATTERY PACK



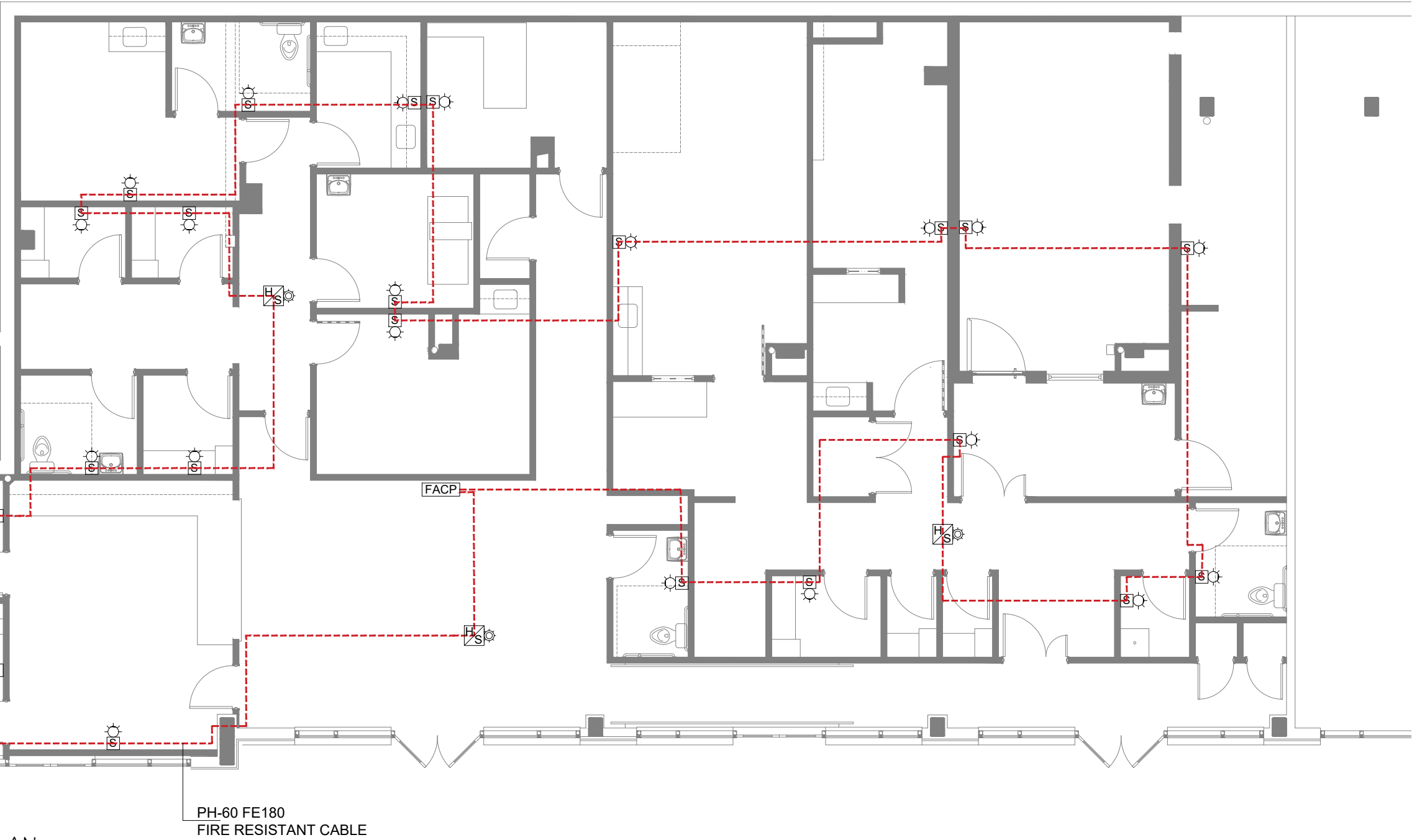
1
A-2.0
FIRE ALARM FLOOR PLAN
SCALE: 3/16" = 1'-0"
GROUND LEVEL

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SCALE: 3/16" = 1'-0"
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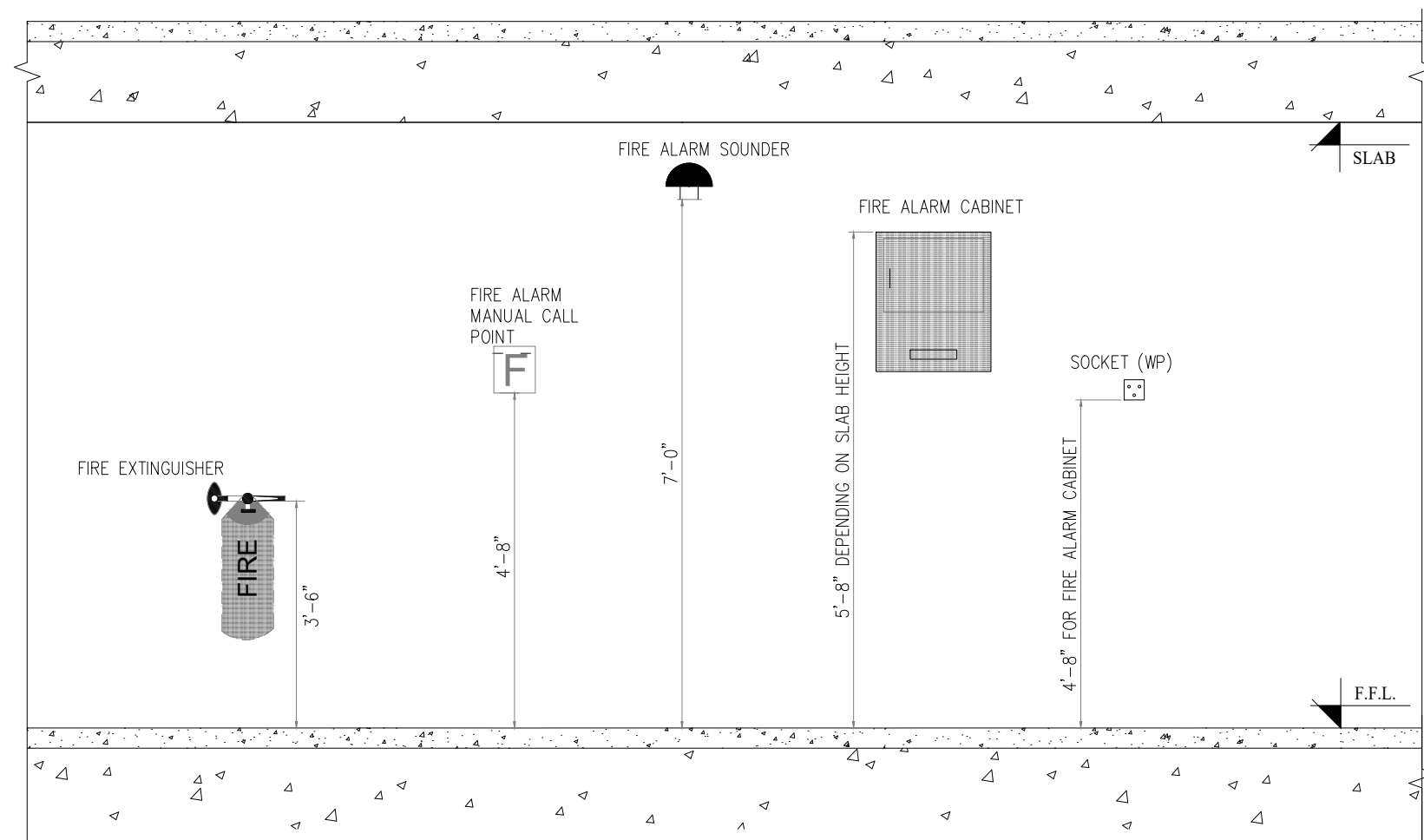
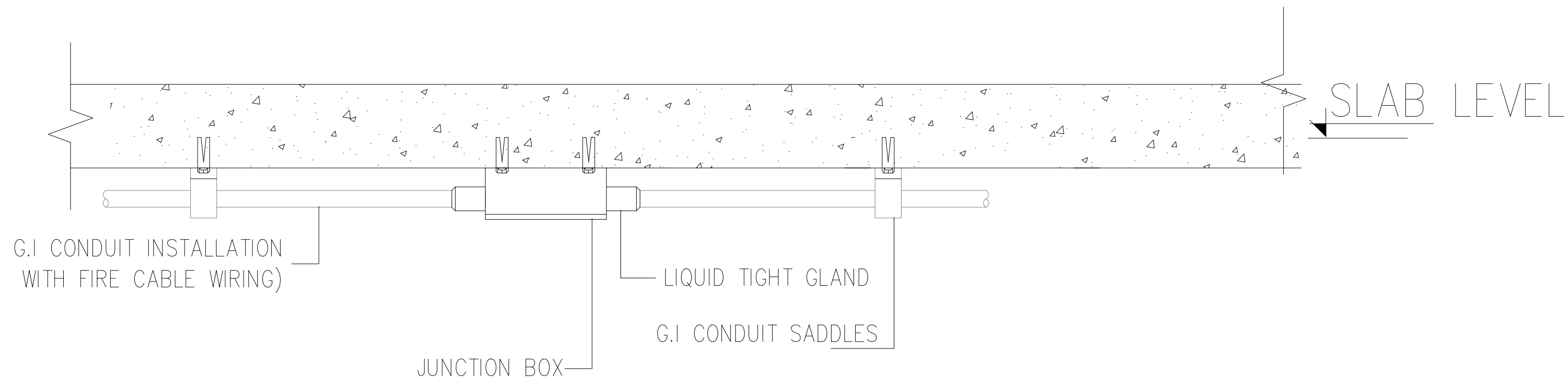


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	CARBON MONOXIDE
	HEAT DETECTOR
	FIRE EXTINGUISHER
	FIRE ALARM CONTROL PANEL
	FIRE ALARM ANNUNCIATION PANEL
	REMOTE EMERGENCY HEAD
	BATTERY PACK



1
A-2.0
EMERGENCY LIGHTING FLOOR PLAN
SCALE: 3/16" = 1'-0"
GROUND LEVEL



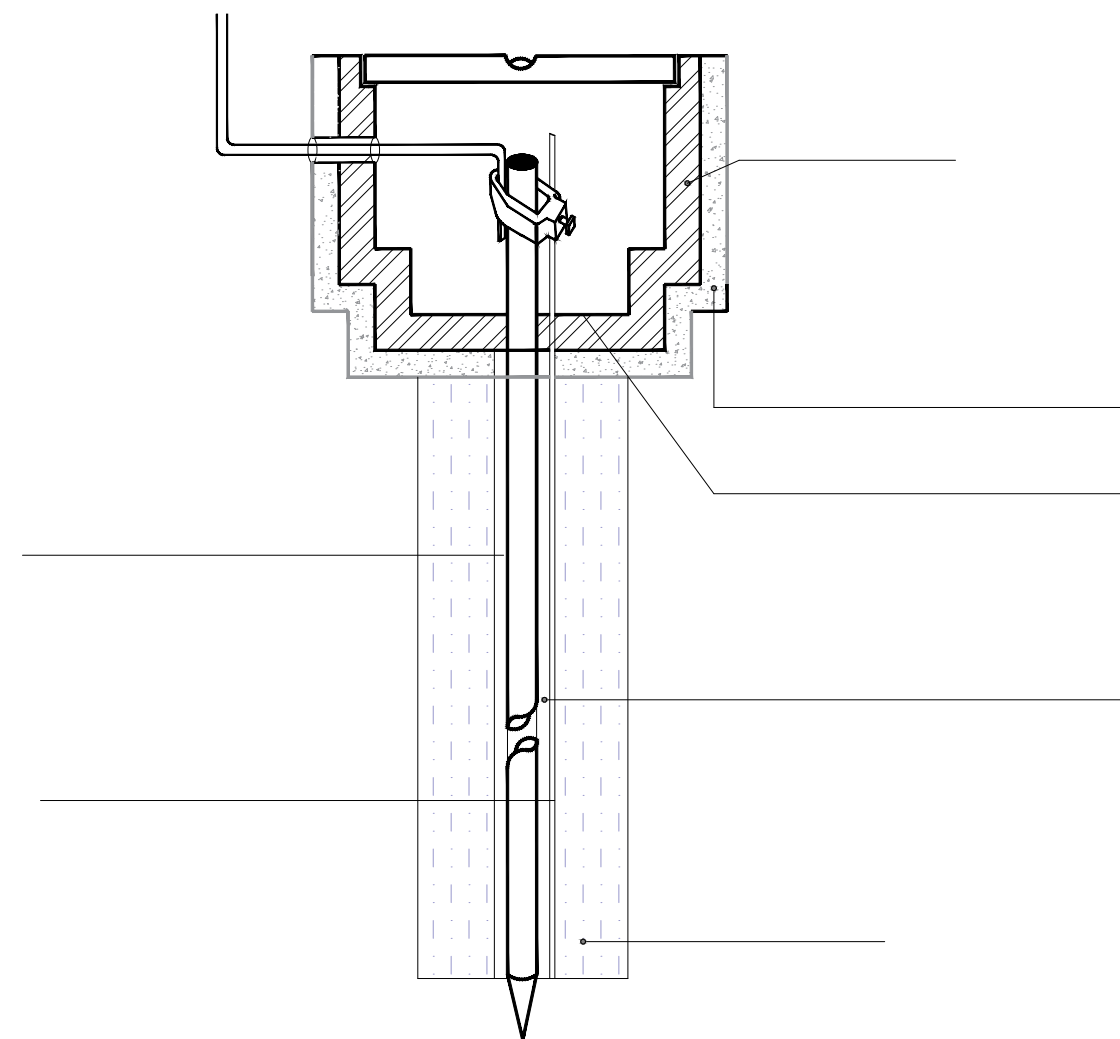


10
EL

STANDARD MOUNTING HEIGHT OF
FIRE ALARM ACCESSORIES

SCALE : N.T.S

TYPICAL DETAIL OF ROD EARTHING



TYPICAL DETAIL OF COPPER ROD

